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ASSIGNMENT - 1

Q.1 Explain the key feature and advantages of using Flutter for mobile app development

→ Flutter is a popular open source UI toolkit by google for building natively compiled apps for mobile (ios & Android) web and desktop from a single codebase

Key features of flutter:

- 1) Single Codebase: Write once, run on multiple platform (ios, Android, web)
- 2) Dart Programming language: Uses Dart, which is optimized for fast performance and ahead of time (AOT) compilation
- 3) Hot Reload: Instantly reflects changes in the app without restarting, making development faster and more efficient
- 4) Rich Widget Library: Provides a vast collection of customizable widgets that support Material

Advantages of Using Flutter!

1. Faster Development Time: Hot reload a single code base reduces effort and time.
2. Cost Effective: Since developers write one codebase for multiple platforms it reduces costs.
3. Consistent UI: Flutter renders everything in its own engine, ensuring uniform look across devices.

Q15] Discuss how the Flutter framework differs from traditional approach and why it has gained popularity?
 as the developers community?

→ Flutter uses a single D.B for multiple platforms, unlike native development that requires separate code for iOS (Swift) & Android (Kotlin). It does not rely on platform specific UI components but instead renders everything using its own engine.

Flutter has gained popularity due to its faster development cost efficient and cross platform support. Business prefer it as it reduces development time & cost while delivering high performance apps.

Describe the concept of the widget tree in flutter. Explain the widget composition is used to build complex UI.

In Flutter, everything is a widget (button, text, layout, etc.). These widget tree determines the UI. The widget composition is used to build complex UI.

Flutter encourages a composition-based approach rather than inheritance.

Instead of creating large, flat are combined to form complex UIs. A column widget can hold multiple Text & Button widget, creating a structured layout.

Qb Provide ex of commonly used widgets & their roles in creating a widget tree.

→ 1) Structural widget

- Scaffold - Provide basic structure of screen.

- Container : Used for styling
- Column & Row : Used for vertical & horizontal

2) Interactive widget.

- Textfield - for user input
- Elevated Button - Clickable buttons

3) Styling widget

- Padding, Ads, spacing around widget
- Align, center, Adjust alignment

4) List & Scrollable widget

- View - Scrollable widget
- GridView - provide display items in grid.

Example: Simple Widget Tree -

```
Scaffold (
  appBar: AppBar (title: Text("Flutter App")),
  body: Column
    children: [
      Text("Welcome to flutter"),
      ElevatedButton(
        onPressed: () {}, child: Text("Click me")
      )
    ]
);
```

Discuss the importance of state management in Flutter application.

Importance of State Management in Flutter
Application state management refers to handling dynamic data that changes over time. The app remains interactive, ensuring proper state management helps in performance optimization, code maintainability & better UI behavior.

Q3 b) Compare & Contrast the different state management in Flutter approaches available in Flutter, provide scenarios where each approach is suitable.

→ Comparison of state management in flutter
Description Suitable scenarios
to update UI, Apps needing global state sharing (eg - user authentication)
• Riverpod more scalable than provides with improved dependency injection & state handling. large, complex apps requiring modularisation & scalable state management (eg - e-commerce app)

4 a) Explain the process of Integrating firebase with a flutter application. Discuss the benefits of using firebase as a backend solution.

→ Integration process:
setup firebase console
Create a firebase project.
Register the App for Android & ios
Download & add Google service - info.plist (ios)
install firebase Dependencies.

yaml

dependencies -

firebase_core: latest-version

firebase_auth: latest-version

cloud_firestore: latest-version

initialize firebase in flutter

dart

void main() async{

T: Lateral (M)

Q46) Highlight the Firebase services commonly in Flutter & provide brief overview of data synchronization is achieved.

→ Commonly used Firebase services in Flutter are:

- Firestore (NoSQL DB) for realtime data sync
- Firebase Authentication (user sign-in with Gmail, Google, Facebook)
- Firebase Analytics (App usage analytics)

Data Synchronization in Firebase Firestore allows realtime data using Snapshot Listeners.

Ex = Of realtime listener in Firestore data.

```
Firestore.instance.collection('my-snapshot').listen((snapshot) {  
  for (doc in snapshot.docs) {  
    print(doc.data());  
  }  
})
```